

Product Requirements Document

AI Tennis Coaching & Player Development Platform

Status: Product Definition / MVP PRD

Version: 2.0

Date: September 12, 2026

Primary Launch Priority: Gametime

Platforms: iOS/Android mobile application + responsive web application

1. Executive Summary

The product is an AI-powered tennis development platform designed primarily for players who want personalized, high-quality feedback but cannot afford frequent access to high-end tennis coaching.

The platform creates an ongoing development loop:

Play → Capture → Analyze → Identify → Learn → Practice → Play Again → Measure Improvement

Its primary differentiator is **Gametime**, a courtside match-recording system that allows parents, coaches, friends, or other approved spectators to record what happens throughout a player's match.

Instead of relying only on the final score, Gametime captures how points were won and lost, optional shot information, observations, important moments, match conditions, and score context.

The platform transforms this information into understandable match statistics, longer-term trends, personalized AI analysis, recommendations, and learning material.

The goal is not to replace human tennis coaches.

The goal is to provide players with the continuous observation, objective measurement, personalized education, and progress tracking that normally require frequent access to expensive coaching.

2. Product Vision

Make personalized tennis development accessible to any motivated player regardless of their ability to afford frequent private coaching.

The platform should gradually understand:

- how the player competes;
- how they win and lose points;
- which weaknesses repeatedly appear;
- which areas are improving;
- their goals;
- their UTR/competitive level;
- what they understand conceptually;
- what coaches and observers notice;
- what they should learn or practice next.

The experience should increasingly feel like:

"This system understands my tennis game."

3. Target Users

3.1 Primary User

The primary user is a tennis player who:

- wants serious, personalized improvement;
- cannot afford frequent high-end coaching;
- wants objective information about their game;
- wants guidance between coaching sessions or without having a regular coach;
- may be competitive or recreational.

Competitive junior players are expected to represent an important segment, but the platform is not restricted to juniors.

3.2 Secondary Users

Parent / Guardian

For junior players, the parent or guardian controls privacy and data-sharing permissions.

Parents may also serve as Gametime scorers.

Players should otherwise have normal freedom to use the learning and development experience.

Coach

Coaches can connect to approved player profiles and view relevant tennis data.

They may:

- review player analytics;
- review match history;
- see goals and trends;
- see scorer observations;
- see AI insights;
- add coaching notes;
- make recommendations.

Coaches cannot directly edit the player's profile or underlying player statistics.

Scorer / Observer

A temporary scorer may be a:

- parent;
- coach;
- friend;
- teammate;
- spectator;
- other trusted person.

Scorers can contribute information to a specific match without receiving access to the player's private profile.

Administrator

Administrators manage educational content, AI-generated lessons, support requests, moderation, and product operations.

4. Product Principles

4.1 Forgiving Data Collection

Gametime must never require perfect recording.

Scorers may:

- skip optional questions;
- miss points;
- leave questions unanswered;
- add or omit notes;
- correct mistakes;
- flag important points;
- contribute different interpretations.

Missing information must be stored as **unknown/not recorded**, not interpreted as "did not happen."

4.2 Evidence Before Conclusions

The AI must distinguish between:

- isolated incidents;
- match-specific patterns;
- recent emerging trends;
- persistent long-term tendencies.

A behavior appearing a few times should not become a permanent statement about the player's game.

The AI should use cautious language when evidence is limited.

4.3 Data Before Advice

Immediately after competition, players primarily want to understand what happened.

Deep coaching recommendations should remain available, but should not overwhelm the immediate post-match experience.

4.4 Explain Recommendations

Personalized recommendations should explain why they appeared.

Example:

Recommended because your backhand unforced-error rate has increased across your recent matches.

4.5 Accessibility

Affordability is central to the product mission.

The product should help players find:

- affordable learning;
- lower-cost tennis courts;
- suitable tournaments;
- community playing opportunities.

Monetization should not recreate the same accessibility problem the product exists to address.

5. MVP Strategy

5.1 Flagship Launch Priority

Gametime is the #1 launch priority.

If engineering or design tradeoffs are required, Gametime and its analytics receive priority.

The central MVP proposition is:

Record what actually happens → understand your game → know what to improve.

5.2 Core Launch

- Gametime
- match scoring
- forgiving point recording
- multiple observers
- match analytics
- critical-point detection
- post-match Wrapped-style recap
- deep match review
- historical analytics
- AI match analysis
- AI Tennis Assistant
- goals
- player profiles
- mobile/web synchronization

5.3 Supporting Launch

- Learn library
- professional-player AI lessons
- quizzes
- Daily Challenge
- drills
- mastery tracking
- personalized educational recommendations
- mental-performance education
- strategy education
- tennis fitness education

5.4 Launch If Feasible

- UTR integration
- tournament recommendations
- affordable court discovery
- coach dashboards
- connection/permission system
- full admin dashboard

5.5 Coming Soon

Player video upload and personalized computer-vision technique analysis.

The Upload interface may be visible at launch but must clearly display **Coming Soon**.

6. Information Architecture

6.1 Home

Home should remain intentionally simple.

Primary greeting:

Hello, [Name]. What would you like to work on today?

The system may surface several personalized starting points based on current needs.

Examples:

- Improve my serve
- Practice positioning
- Continue recommended lesson
- Take today's challenge
- Review recent match

Home should not become a dense analytics dashboard.

6.2 Primary Navigation

Recommended primary navigation:

- Home
- Learn
- My Tennis / Data
- Gametime
- Profile

The **AI Assistant** remains accessible through a persistent floating action/button.

6.3 Web

The web experience is a core product surface.

It is particularly optimized for:

- longer learning sessions;
- professional video lessons;
- quizzes;
- deep analytics;
- historical match exploration;
- coaches;
- administration.

6.4 Mobile

Mobile is optimized for:

- Gametime;
- courtside usage;
- Daily Challenge;
- quick lessons;
- notifications;
- AI Assistant;
- quick analytics review.

7. Cross-Device Continuity

Mobile and web use the same account and backend.

The following synchronize automatically:

- match history;
- statistics;
- lesson progress;
- current lesson position;
- quiz progress/results;
- mastery;
- Saved content;

- goals;
- recommendations;
- coach information;
- AI insights.

A player should be able to begin a lesson on a computer and continue it on a phone.

8. Player Onboarding

Onboarding should gather enough information to personalize the experience without becoming excessively long.

Potential information includes:

- name;
- age;
- approximate location;
- UTR;
- dominant hand;
- one-handed/two-handed backhand;
- years playing;
- competitive/recreational status;
- primary tennis goals;
- sub-goals;
- self-identified weaknesses.

Location should collect only the precision necessary for recommendations.

Continuous precise location is not required.

9. Goals

Players may establish:

Main Goals

Examples:

- reach a target UTR;
- make a school team;
- prepare for college tennis;
- improve overall competitive results.

Sub-Goals

Examples:

- improve second serve;
- reduce unforced errors;
- improve backhand positioning;
- compete more frequently.

Goals are active inputs to personalization.

The AI may use:

- goals;
- UTR;
- match performance;
- recent trends;
- quiz/mastery data;
- accepted coach recommendations

to maintain a recommended development plan.

10. UTR Integration

The product should investigate integration with the **UTR Sports Engage API**.

Desired information, subject to actual API capabilities and licensing, includes:

- player identity;
- current UTR;
- historical UTR;
- match history;
- opponents;

- scores;
- dates;
- events.

If supported, users should be able to connect/import their available UTR history.

UTR-imported matches and Gametime matches must remain distinguishable.

An imported match may provide:

- opponent;
- result;
- score;
- date;
- rating context.

It must never be represented as containing point-level statistics that were never collected.

Manual UTR entry remains a fallback.

11. Gametime

11.1 Purpose

Gametime enables approved spectators to record what happens during a tennis match without requiring the player to interact with a device during play.

It should prioritize:

speed + flexibility + forgiveness.

12. Match Setup

Before a match, Gametime may capture:

- player;
- opponent;

- opponent UTR/profile where available;
- tournament/event;
- date;
- start time;
- location;
- venue;
- indoor/outdoor;
- surface;
- weather;
- match format;
- scoring format;
- who serves first;
- starting court side.

Date/time should be automatically populated where possible.

Weather/location information may be automatically populated when appropriate and permitted.

13. Temporary Scorer Access

A player/parent can generate a temporary:

- QR code;
- match invitation;
- match code.

A scorer joining through this mechanism receives access only to that match.

They may:

- record points;
- answer structured questions;
- add notes;
- dictate notes;
- flag points;
- submit observations.

They may not access:

- historical player data;
- goals;

- private analytics;
- AI conversations;
- profile information;
- unrelated matches.

Temporary access ends after the match unless otherwise authorized.

14. Multiple Contributors

Multiple approved people may contribute to the same match.

Contributors may disagree.

The system should preserve differing observations rather than silently deciding which person is correct.

After the match, the player can resolve meaningful conflicts.

Unresolved disagreements should be marked uncertain and treated appropriately by analytics.

15. Point Recording

After each point, Gametime asks lightweight structured questions.

The exact question tree should be optimized through usability testing.

Core information includes:

Who won?

- Player
- Opponent

How did the point end?

Examples:

- Ace

- Winner
- Forced Error
- Unforced Error
- Double Fault
- Opponent Winner
- Opponent Forced Error

Shot

Where relevant:

- Forehand
- Backhand
- Serve
- Volley
- Slice
- Other

Not every question is mandatory.

Additional detail may be placed into notes rather than requiring excessive structured input.

16. Notes

Notes are completely optional.

16.1 Point Notes

Short and specific.

Example:

"Late getting to backhand."

16.2 Game Notes

Broader observations about the game.

16.3 Set Notes

Patterns across the set.

16.4 Match Notes

High-level observations about the match overall.

Notes may be typed or dictated through voice-to-text.

The scorer should be able to correct transcription mistakes.

17. Editing & Corrections

Scorers can:

- undo;
- edit recent points;
- correct classifications;
- modify notes.

After submission, authorized users can correct obvious errors.

Analytics should use corrected information.

Material edits should maintain a lightweight audit history.

18. Critical Points

18.1 Manual Flags

Observers may flag a point as:

- Important
- Critical

- Worth reviewing

18.2 System Detection

Gametime understands live score context.

The system can recognize potentially high-pressure situations and prompt:

"This looks like a high-pressure point. Would you like to flag it?"

The scorer can:

- accept;
- decline;
- ignore.

The system should consider score context when identifying important moments.

Examples include:

- break points;
- set points;
- match points;
- tiebreak situations;
- other consequential score states.

The AI determines which moments deserve prominence in the final recap based primarily on match context and recorded information.

19. Offline Gametime

Gametime must support offline match recording.

While offline:

- scoring continues;
- structured responses work;
- notes work;
- flags work;
- match information is stored locally.

When the scorer taps Submit while offline:

1. Match is validated locally.
2. Match is saved on the device.
3. User sees a clear "Waiting to sync" state.
4. When connectivity returns, data automatically uploads.
5. Cloud analytics/AI processing begins.

Offline mode must not risk losing an entire recorded match because connectivity disappeared.

20. Immediate Post-Match Experience

The immediate post-match experience should feel like **Spotify Wrapped for a tennis match**.

It should be:

- visual;
- quick;
- swipeable;
- understandable;
- engaging;
- data-first.

It is not the place for an enormous coaching report.

Potential recap cards include:

- final score;
- match duration;
- set results;
- winners;
- unforced errors;
- forced errors;
- aces;
- double faults;
- shot breakdown;
- strongest set;
- pressure-point performance;
- critical moments;
- major differences between sets;

- concise AI observations.

Raw statistics that can be computed immediately may appear first.

After cloud/AI processing completes, the complete recap becomes available and the player receives:

"Your match report is ready."

21. Player Reflection

After the match, the player may optionally add their own reflection.

Possible prompts:

- How did you feel you played?
- What felt strongest?
- What felt difficult?
- Anything you want to remember?

Reflection may be typed or dictated.

Player reflection remains distinguishable from observer notes.

If subjective perception conflicts with recorded statistics, the AI may respectfully surface the discrepancy rather than automatically choosing one as correct.

22. Deep Match Analysis

Later, the player can deliberately enter Match Details for deeper analysis.

Available exploration includes:

- sets;
- games;
- point outcomes;
- shot categories;
- error categories;

- pressure points;
- notes;
- observer disagreements;
- comparisons;
- AI analysis.

The full chronological point timeline remains available but is not the default experience.

Default review prioritizes critical and meaningful moments.

23. Overall Player Analytics

The My Tennis/Data section combines multiple matches.

Players may analyze:

- all time;
- season;
- last N matches;
- custom period;
- manually selected matches.

Analytics should support drill-down from broad trends into specific categories.

Examples:

- overall unforced errors;
- backhand unforced errors;
- serve performance;
- ace trends;
- double-fault trends;
- performance by set;
- pressure-point performance;
- performance by surface;
- indoor vs. outdoor;
- weather context;
- time-of-day context;
- opponent history.

The system must avoid drawing strong conclusions from insufficient samples.

24. Head-to-Head History

When the same opponent appears multiple times, the platform creates a head-to-head history.

It may show:

- W/L record;
- previous scores;
- UTR context;
- changes between meetings;
- player performance patterns.

Analysis must focus on the user's own recorded performance rather than inventing opponent characteristics.

25. AI Match Analysis

AI analysis combines relevant information such as:

- structured point data;
- score context;
- set/game context;
- scorer notes;
- player reflections;
- historical trends;
- goals;
- UTR;
- accepted coach recommendations.

AI conclusions must be evidence-weighted.

Isolated evidence

Keep within match-specific analysis.

Repeated evidence

May become a broader insight.

Recent repeated evidence

May be labeled as an emerging trend.

Long-term repeated evidence

May contribute to the player's overall tennis profile.

26. Personalized Recommendations

When the system detects a meaningful weakness, it should close the learning loop.

Example:

Backhand weakness detected → relevant drills + lesson + quiz recommended.

Recommendations should identify their evidence where possible.

Players may be encouraged to practice a skill for multiple sessions before reassessment.

For example:

"We recommend practicing these backhand drills for at least five sessions before reassessing."

This is guidance, not enforcement.

Players do not need to prove that they completed the sessions.

27. AI Tennis Assistant

A persistent AI Tennis Assistant is available throughout the product.

The assistant must have permissioned access to relevant player tennis data.

Depending on the question, it may retrieve:

- Gametime data;
- historical statistics;
- goals;
- UTR;
- trends;
- scorer notes;
- player reflections;
- lesson performance;
- quiz/mastery information;
- coach recommendations;
- future video-analysis outputs.

Example questions:

- What should I work on?
- Has my serve improved?
- Why am I losing more backhand points?
- What improved this month?
- What should I practice before my next tournament?
- Explain a kick serve.

The system should retrieve only the relevant data required to answer the question.

Persistent storage of complete AI conversation history is not an MVP requirement.

Players can instead explicitly request historical comparisons.

28. AI Feedback

AI-generated insights should support:

- Helpful
- Inaccurate / Not Helpful
- Optional explanation

Feedback should help administrators identify quality problems.

A negative rating must not automatically alter the underlying player data.

29. Learn

Learn contains structured tennis education.

Major categories include:

1. Positioning
 2. Shots
 3. Serves
 4. Strategy & Tactics
 5. Mental Performance
 6. Tennis Fitness & Conditioning
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30. Positioning

Positioning focuses on getting into an effective position relative to the ball.

Topics may include:

- movement to the ball;
- spacing;
- preparation;
- recovery;
- open stance;
- semi-closed stance;
- closed stance;
- situational footwork.

Content includes:

- professional examples;
 - lessons;
 - quizzes;
 - drills.
-

31. Shots

Shot categories include:

Forehand

Technique and variations.

Backhand

One-handed and two-handed where appropriate.

Volley

Forehand and backhand volleys within the lesson structure.

Slice

Forehand and backhand slices.

Other shot categories may be added over time.

32. Serves

Serve education receives deeper organization.

Serve types may include:

- Flat
- Kick
- Slice
- Other variations

Each serve type should cover:

- deuce-side serving;
- ad-side serving;
- positioning;

- technique;
 - placement;
 - tactical use.
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33. Strategy & Tactics

Topics may include:

- point construction;
 - attacking vs. defending;
 - shot selection;
 - serve + 1;
 - return strategy;
 - approaching the net;
 - adapting during matches;
 - playing different styles;
 - court positioning;
 - patterns of play.
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34. Mental Performance

Educational topics may include:

- between-point routines;
- resetting after mistakes;
- focus;
- confidence;
- handling pressure;
- maintaining composure;
- decision-making under pressure.

The product should not make psychological diagnoses based on tennis behavior.

35. Tennis Fitness & Conditioning

Educational material may cover age-appropriate:

- warm-ups;
- mobility;
- agility;
- tennis-specific endurance;
- footwork conditioning;
- recovery;
- general athletic preparation.

The platform should avoid extreme training recommendations or restrictive diet/body guidance, particularly for junior players.

36. Learn From Professionals

Professional-player clips become structured AI-assisted lessons.

The purpose is not simply to show a professional playing well.

The system should explain:

- what the professional is doing;
- how they move toward the ball;
- how they prepare;
- positioning relative to the ball;
- stance;
- footwork;
- body movement;
- contact preparation;
- recovery;
- tactical context where observable;
- why these actions are effective;
- what the learner can apply.

AI must not claim measurements or details that cannot reasonably be inferred from the footage.

37. Professional Lesson Content Pipeline

Professional lessons should be generated once, not re-analyzed every time a player watches.

Workflow:

1. Admin adds licensed/authorized professional footage.
2. AI analyzes the clip.
3. AI drafts structured educational content.
4. Admin reviews.
5. Admin edits where necessary.
6. Admin approves.
7. Lesson is published.
8. The same base lesson is reused.

Human approval is mandatory before publication.

Personalization occurs primarily through recommendation selection rather than regenerating the entire lesson for every user.

38. Quizzes

Quizzes should emphasize applied understanding.

Questions may use:

- images;
- diagrams;
- short clips;
- tennis scenarios.

Players may be asked:

- what should happen next;
- what is incorrect;
- where the player should move;
- what shot is appropriate;

- what tactical decision is best.

The AI explains why the answer was strong, incomplete, or incorrect.

Quizzes should test understanding, not merely memorization.

39. Mastery

The system tracks knowledge mastery by topic.

A weak quiz performance may trigger:

- lesson review;
- additional questions;
- related recommendations.

Future questions should test the same concept using different situations.

Consistently strong performance may move a topic toward **Mastered**.

Knowledge mastery and actual match execution must remain separate concepts.

A player knowing backhand theory does not prove they execute the backhand effectively during competition.

40. Mastery Map

Players may access a visual learning-progress map.

Example:

Positioning

- Open stance — 85%
- Semi-closed — 72%
- Closed — 90%

Shots

- Forehand — 81%
- Backhand — 64%
- Volley — 76%

Serves

- Flat — 88%
- Slice — 71%
- Kick — 55%

This is a secondary progress feature rather than the primary Home experience.

41. Daily Challenge

Daily Challenge is a scenario-based tennis learning experience.

Typical structure:

1. Show an image or very short video.
2. Stop before the outcome.
3. Ask the player what should happen next.
4. Player usually responds in writing.
5. AI evaluates the reasoning.
6. AI explains the situation.
7. Player may ask follow-up questions.

Daily Challenge difficulty should adapt to player level.

Higher-level players should receive more nuanced scenarios.

42. Gamification

Daily Challenge and learning may include:

- streaks;
- XP;
- levels;
- achievements;

- leaderboards.

Gamification should reward:

- consistency;
- participation;
- learning;
- improvement.

It should not reward only perfect answers.

Junior privacy must be considered when designing leaderboards.

43. Search

Learn includes universal search.

Players may search topics such as:

- kick serve;
- open stance;
- returning a fast serve;
- playing against defensive players.

Search spans:

- lessons;
- professional breakdowns;
- quizzes;
- drills;
- strategy;
- mental content;
- fitness content.

Potential filters include:

- category;
- content type;
- topic;
- difficulty;
- UTR/level.

44. Saved Content

Players may bookmark:

- lessons;
- professional clips;
- drills;
- quizzes;
- Daily Challenges.

Saved content synchronizes across web and mobile.

45. Pre-Match Brief

Players may receive a concise pre-match experience.

It may surface:

- current goals;
- recent areas of focus;
- relevant head-to-head history;
- concise tactical reminders;
- recent work.

The system should avoid introducing major technical changes immediately before competition.

46. Coach Dashboard

Coach accounts can maintain a roster of linked players.

The coach can select a player and view approved information including:

- match history;
- analytics;

- UTR;
- trends;
- goals;
- scorer notes;
- AI insights;
- learning progress;
- future video-analysis results.

The dashboard should make switching between players fast.

47. Coach Recommendations

Coaches may:

- leave notes;
- submit recommendations.

They may not directly edit:

- player profile;
- statistics;
- underlying match data;
- player settings.

When a coach submits a recommendation:

1. Player reviews it.
2. Player accepts or declines.
3. If accepted, it becomes an input to personalization.
4. AI may recommend relevant lessons, quizzes, videos, or drills.

This creates:

Coach judgment → Player approval → AI follow-through

48. Connections & Permissions

Players may connect with other people.

Connections do not automatically grant complete data access.

Permission levels determine what each connection can see.

Examples:

- friend sees limited progress/streak information;
- coach receives deeper approved access;
- scorer receives match-only access;
- parent/guardian controls junior sharing permissions.

No junior player's private data should be public by default.

49. Parent / Guardian Permissions

For users below the applicable age of independent consent:

- parent/guardian approval is required;
- parent controls data-sharing permissions;
- parent approves or declines coach access;
- parent manages sensitive privacy/account decisions.

After appropriate consent is established, the player can freely use ordinary product functionality.

50. Human Support / Ask the Team

The platform includes a channel for users to contact the product team.

This is separate from the AI Tennis Assistant.

Use cases include:

- AI answer was unsatisfactory;
- user believes analysis is incorrect;
- product questions;
- account/support issues;

- questions requiring human judgment.

With appropriate permissions, authorized support personnel may review relevant player data to provide a useful answer.

Access to junior data must be tightly controlled.

51. Admin Dashboard

Admins require a dedicated management interface.

Functions include:

- upload/manage professional clips;
 - review AI-generated lessons;
 - edit lessons;
 - approve/publish lessons;
 - create/manage quizzes;
 - create/manage Daily Challenges;
 - manage Practice Library;
 - answer human-support questions;
 - manage reported issues;
 - review AI-quality feedback;
 - view high-level usage analytics;
 - manage content taxonomy.
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52. Courts & Playing Opportunities

The platform should help users locate tennis opportunities, with affordability prioritized.

Recommendations may include:

- free public courts;
- low-cost courts;
- community tennis facilities;
- affordable playing opportunities.

Approximate location should be sufficient for most recommendations.

53. Tournament Recommendations

Tournament discovery should consider:

- player UTR;
- accepted rating range;
- age eligibility;
- division;
- location;
- travel distance;
- entry cost;
- date;
- registration availability where known.

Affordable and appropriately competitive tournaments should receive priority.

The initial version can keep tournament recommendations simple rather than requiring a detailed explanation page.

54. Notifications

MVP notification categories:

- suitable upcoming tournaments;
- completed match reports;
- new lessons relevant to a demonstrated weakness;
- reminders after meaningful periods without playing/logging a match.

Users can control notification categories.

Generic engagement spam should be avoided.

55. Future Player Video Analysis

Status: Coming Soon

Players will eventually upload a limited number of videos based on subscription tier.

The intended future system will analyze:

- body positioning;
- joint relationships;
- movement;
- preparation;
- positioning relative to the ball;
- technique patterns;
- differences across uploads.

The system should learn the player's typical playing mechanics over time while also comparing relevant technique against evidence-based tennis mechanics.

The feature should not force every player into one identical professional style.

This capability requires separate technical validation and significantly more resources.

It is not required for initial launch.

56. Future Video Storage Principle

When video analysis launches, follow:

Process → Extract → Delete

Prefer retaining:

- derived measurements;
- feedback;
- small annotated references;
- player shot-profile data

instead of indefinitely retaining raw video.

This is particularly important because many users will be minors.

57. Monetization

Exact pricing and subscription tiers remain **TBD** until:

- product scope stabilizes;
- AI usage is measured;
- infrastructure costs are understood;
- willingness-to-pay is tested.

Primary principle:

Features requiring greater marginal AI/computing resources may require higher subscription levels.

However, the free experience must remain useful enough to support the accessibility mission.

Future video-analysis volume is a natural subscription differentiator.

58. Data Model — High-Level Entities

Core entities include:

- User
- Player Profile
- Parent/Guardian
- Coach
- Connection
- Permission
- Goal
- Sub-Goal
- Match
- Set
- Game
- Point

- Point Classification
 - Observer
 - Observation
 - Critical Point Flag
 - Player Reflection
 - Match Context
 - Opponent
 - UTR Record
 - AI Insight
 - Recommendation
 - Lesson
 - Professional Clip
 - Quiz
 - Quiz Attempt
 - Mastery Topic
 - Drill
 - Daily Challenge
 - Saved Content
 - Tournament
 - Court/Facility
 - Support Request
 - AI Feedback
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59. AI Architecture Principles

AI should not receive the player's entire history for every request.

Instead:

1. Understand user question/task.
2. Query relevant structured information.
3. Calculate necessary statistics.
4. Retrieve relevant observations.
5. Provide the relevant context to the AI.
6. Generate explanation/recommendation.

This reduces:

- token usage;
- latency;

- AI cost;
- unnecessary data exposure.

Structured data remains the source of truth.

AI primarily interprets and communicates that data.

60. Privacy & Youth Safety

Because junior players are an important audience, privacy must be designed into the system.

Requirements include:

- appropriate parental consent;
- private-by-default junior accounts;
- explicit connection permissions;
- revocable coach access;
- limited scorer permissions;
- role-based admin access;
- no public raw junior match/video data by default;
- secure storage;
- clear retention policies;
- data minimization;
- deletion/account-management controls.

Exact requirements must be reviewed against applicable laws and jurisdictions before launch.

61. AI Safety & Quality

AI-generated tennis guidance should:

- distinguish evidence from inference;
- avoid overconfidence;
- avoid diagnosing physical/psychological conditions;
- acknowledge insufficient data;
- avoid treating isolated incidents as permanent weaknesses;
- distinguish knowledge mastery from physical execution;

- clearly identify uncertainty;
- support user feedback.

Published educational lessons generated by AI require human approval.

62. Professional Footage Rights

Professional tennis footage must be legally authorized for use.

The product should not assume arbitrary tournament broadcasts, social-media clips, or copyrighted professional footage may be republished.

Content operations must establish appropriate:

- licensing;
 - partnerships;
 - permissions;
 - owned footage;
 - other lawful sources.
-

63. Non-Goals for Initial Release

Initial release does not aim to provide:

- working personalized video-upload biomechanics;
 - injury diagnosis;
 - medical advice;
 - complete replacement for human coaching;
 - live player-side coaching during points;
 - direct player-vs-player quiz battles;
 - permanent AI conversation memory;
 - unrestricted public junior profiles;
 - perfect automatic interpretation from incomplete data.
-

64. Success Metrics

Success should be evaluated across player outcomes, product behavior, accessibility, and business health.

Player Improvement

- change in relevant match-performance metrics;
- progress toward player goals;
- improvement in recurring weaknesses;
- UTR progression where appropriate.

Gametime Adoption

- matches recorded per active player;
- percentage of users recording multiple matches;
- completion rate;
- scorer retention;
- percentage of matches with useful structured data.

Insight Quality

- percentage of AI insights marked helpful;
- recommendation engagement;
- frequency of inaccurate/irrelevant reports;
- player/coach satisfaction.

Learning

- recommended lessons started;
- lesson completion;
- quiz engagement;
- mastery progression;
- Daily Challenge participation/streaks.

Behavioral Impact

Measure whether players actually use analytics to change what they practice.

Accessibility

Measure adoption among players who do not have frequent access to private coaching.

Retention

- weekly active users;
- monthly active users;
- returning Gametime users;
- cohort retention.

Coach Adoption

- active coaches;
- linked players per coach;
- coach recommendations;
- player acceptance/engagement with recommendations.

Business

- free-to-paid conversion;
 - subscription retention;
 - AI cost per active user;
 - gross margin as AI features expand.
-

65. Key Risks

Scorer Burden

Even forgiving recording requires attention.

Mitigation: optional questions, large controls, voice notes, skip behavior, usability testing.

Data Quality

Different observers may classify points differently.

Mitigation: preserve uncertainty, support corrections, allow player resolution.

AI Overgeneralization

Sparse data may produce misleading conclusions.

Mitigation: evidence thresholds, confidence-aware language, explicit trend logic.

AI Cost

Personalized analysis can become expensive.

Mitigation: structured-data retrieval, cached/pre-generated educational content, usage-aware subscriptions.

Junior Privacy

Large junior audience creates additional responsibility.

Mitigation: guardian consent, permission architecture, privacy-by-default.

UTR Dependency

Desired API functionality may not match available commercial access.

Mitigation: manual UTR entry and independent Gametime database.

Content Rights

Professional footage may be expensive or difficult to license.

Mitigation: content partnerships and legally authorized footage strategy.

Scope

The full vision is large.

Mitigation: Gametime remains the flagship launch priority.

66. Open Questions

The following should be resolved through technical discovery, user testing, legal review, and business validation:

1. Exact Gametime point-entry interaction.
 2. Minimum structured questions per point.
 3. UTR Engage API endpoint availability.
 4. UTR partnership/licensing requirements.
 5. Tournament-data provider.
 6. Court/facility-data provider.
 7. Professional-footage licensing strategy.
 8. AI model/provider architecture.
 9. Exact AI trend thresholds.
 10. Subscription pricing.
 11. Subscription usage limits.
 12. Free-tier boundaries.
 13. Exact junior-consent implementation by jurisdiction.
 14. Weather-data provider.
 15. Voice-transcription implementation.
 16. Offline synchronization conflict handling.
 17. Analytics visualization system.
 18. Professional lesson content-review workflow.
 19. Future CV/video technical feasibility.
-

67. Recommended Build Sequence

Phase 1 — Gametime Foundation

Build:

- authentication;
- player profiles;
- parent permissions;
- match creation;
- scoring engine;
- point classifications;
- notes;
- voice notes;
- critical flags;
- multiple contributors;
- offline storage;
- synchronization;
- basic match statistics.

Primary validation: Can spectators consistently capture useful match information without Gametime becoming annoying?

Phase 2 — Match Intelligence

Build:

- Wrapped-style recap;
- historical analytics;
- trend engine;
- critical-point analysis;
- player reflections;
- AI Match Review;
- AI Tennis Assistant;
- recommendations.

Primary validation: Does the resulting analysis actually help players understand what to improve?

Phase 3 — Learning Loop

Build:

- Learn library;
- professional lessons;
- quizzes;
- drills;
- Daily Challenge;

- mastery;
- Saved;
- search;
- recommendations connecting Gametime → Learn.

Primary validation: Do players act on identified weaknesses?

Phase 4 — Ecosystem

Build/expand:

- coach dashboard;
- coach recommendations;
- connections;
- tournament discovery;
- affordable court discovery;
- UTR integration;
- expanded admin tooling.

Phase 5 — Personalized Video Analysis

After separate technical validation:

- player video upload;
- pose estimation;
- movement analysis;
- player technique profiles;
- longitudinal video comparison;
- extracted-data storage;
- subscription-based upload limits.

68. North-Star Product Experience

A junior player finishes a match.

Their parent and coach have contributed observations through Gametime without needing perfect data entry.

The match syncs.

The player receives a visual, enjoyable recap showing what actually happened.

Later at home, the player opens the website.

They explore deeper analytics and discover that a weakness has appeared consistently across recent matches.

The AI explains the evidence rather than simply declaring that the player is bad at something.

The platform recommends a relevant professional lesson, quiz, and practice drill.

The player's coach may add their own recommendation.

The player practices.

They compete again.

Gametime captures the next performance.

The system determines whether the trend is actually improving.

Over time, the player develops a detailed, objective history of their tennis game without requiring a private coach to watch and remember every point of every match.

That continuous feedback loop is the core product.

69. Product Positioning

This product is not simply:

an AI tennis chatbot.

It is not simply:

a tennis course library.

It is not simply:

a scorekeeping application.

It is a **personal tennis development system built around the player's own data.**

Its strongest differentiation is the connection between:

real match behavior + human observation + objective analytics + AI interpretation + personalized learning + measurable improvement.

That connection should guide future product decisions.